Challenge 8: Ryser's Hypergraph Conjecture

Definition 1 (r-partite)**.** Given $r \in \mathbb{N}$, an r-uniform hypergraph $\mathcal{H}$ is *r-partite* if there is a partition

$$V(\mathcal{H}) = \bigcup_{i=1}^{r} P_i$$

of the vertex set of $\mathcal{H}$ such that $|P_i \cap e| = 1$, for each $e \in E(\mathcal{H})$ and $i \in [r]$.

Definition 2 (Cover number)**.** Given a hypergraph $\mathcal{H}$, a set C of vertices of $\mathcal{H}$ is a *vertex cover* if it contains a vertex of every edge. The smallest cardinality of a vertex cover of $\mathcal{H}$ is called the *cover number* of $\mathcal{H}$. It is denoted by $\tau(\mathcal{H})$.

Definition 3 (Matching number)**.** The *matching number* of a hypergraph $\mathcal{H}$ is the largest cardinality of a set of pairwise disjoint edges of $\mathcal{H}$. It is denoted by $\nu(\mathcal{H})$.

Conjecture 1.1 (Ryser's Conjecture [1])**.** Let $r \in \mathbb{N}$. Every r-partite r-uniform hypergraph $\mathcal{H}$ satisfies:

$$\tau(\mathcal{H}) \leq (r-1) \cdot \nu(\mathcal{H})$$

For $r = 2$, it is a classical theorem of Kőnig, and for $r = 3$ it was proved by Aharoni [2] using the topological method.

References

[1] A. Bishnoi et al. *Ryser's Conjecture for t-intersecting hypergraphs*. 2020. arXiv: 2001.04132 [math.CO]. URL: https://arxiv.org/abs/2001.04132.

[2] R. Aharoni. "Ryser's conjecture for tripartite 3-graphs". In: *Combinatorica* 21.1 (2001), pp. 1–4.